

Sangyeon Yoon

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Research Interest

I am interested in how post-training shapes the behavior of large language models and how to make increasingly capable models safer and more reliable. Here are some questions I am particularly interested in:

- **Post-training and alignment:** How do reinforcement learning shape model behavior? When do they lead to unintended behaviors such as reward hacking, and how can we prevent these failures?
- **Evaluation and oversight:** How can we distinguish genuine improvements from exploitation of imperfect objectives? How can evaluators and monitors remain reliable under optimization pressure?
- **Reasoning and safety:** How does reasoning interact with post-training and alignment? When does it improve safety and oversight, and when does it introduce new failure modes?

Education

Yonsei University

Sep 2025 – Present

Ph.D. in Artificial Intelligence

Hongik University

Mar 2019 – Aug 2025

B.S. in Computer Engineering

- Includes a 2-year mandatory military service in South Korea.

Research Experience

AI-ISL Lab, Yonsei University

Mar 2024 – Present

Graduate Researcher (under **Prof. Albert No**)

EXAONE Lab, LG AI Research

Sep 2025 – Feb 2026

Research Intern

- Contributed to foundation model development, including EXAONE-4.5 and K-EXAONE-236B-A23B.
- Contributed to the post-training team by designing synthetic data for reasoning.

Publications

Preprints

1. Diagnosing and Correcting Reward Hacking in Verifiable Instruction Following
Sangyeon Yoon*, Wooseok Seo*, Hamish Ivison, Hyesoo Hong, Donghoon Kim, Wonjun Kang, Albert No, Jaehyung Kim
Under Review
2. Instruction Following with Composable Constraints
Wooseok Seo, **Sangyeon Yoon**, Valentina Pyatkin, Heuiyeon Yeen, Sunkyoung Kim, Yongil Kim, Youngjae Yu, Seokhee Hong
Under Review

Conference & Workshop Papers

1. Few-Shot Truly Benign DPO Attack for Jailbreaking LLMs
Sangyeon Yoon*, Wonje Jeung*, Yoonjun Cho, Dongjae Jeon, Albert No
Conference on Neural Information Processing Systems (NeurIPS 2026) [Link] [🔗](#)
2. VLMs Trace Without Tracking: Diagnosing Failures in Visual Path Following
Hyesoo Hong, Min Soo Kim, Wonje Jeung, **Sangyeon Yoon**, Dongjae Jeon, Albert No

Conference on Neural Information Processing Systems (NeurIPS 2026) [\[Link\]](#) [🔗](#)

3. BenchPreS: A Benchmark for Context-Aware Personalized Preference Selectivity of Persistent-Memory LLMs
Sangyeon Yoon, Hyesoo Hong, Wonje Jeung, Yongil Kim, Wooseok Seo, Heuiyeen Yeen, Albert No, Sunkyoung Kim
Conference on Empirical Methods in Natural Language Processing (EMNLP 2026 Oral) [\[Link\]](#) [🔗](#)
Previously at SCALE@ICML2026 [Oral (5.8%)]
4. RoboRMBench: Paraphrase Fragility in Vision Language Reward Models
Wonje Jeung, **Sangyeon Yoon**, Hyesoo Hong, Yoonjun Cho, Dongjae Jeon, Bumjun Kim, Jean Oh, Youngjae Yu, Albert No
Conference on Empirical Methods in Natural Language Processing (EMNLP 2026 Oral)
5. Position: The Term “Machine Unlearning” Is Overused in LLMs
Sangyeon Yoon*, Yeachan Jun*, Albert No
International Conference on Machine Learning (ICML 2026 Position Track)
6. DUSK: Do Not Unlearn Shared Knowledge
Wonje Jeung*, **Sangyeon Yoon***, Hyesoo Hong*, Soeun Kim, Seungju Han, Youngjae Yu, Albert No
Findings of the Association for Computational Linguistics (ACL Findings 2026)
7. Rethinking Benign Relearning: Syntax as the Hidden Driver of Unlearning Failures
Sangyeon Yoon, Hyesoo Hong, Wonje Jeung, Albert No
International Conference on Learning Representations (ICLR 2026) [\[Link\]](#) [🔗](#)
Selected as a BK21 Academy Research Fellow
8. A2D: Any-Order, Any-Step Safety Alignment for Diffusion Language Models
Wonje Jeung*, **Sangyeon Yoon***, Yoonjun Cho, Dongjae Jeon, Sangwoo Shin, Hyesoo Hong, Albert No
International Conference on Learning Representations (ICLR 2026) [\[Link\]](#) [🔗](#)
9. SAFEPath: Preventing Harmful Reasoning in Chain-of-Thought via Early Alignment
Wonje Jeung, **Sangyeon Yoon**, Minsuk Kang, Albert No
Conference on Neural Information Processing Systems (NeurIPS 2025) [\[Link\]](#) [🔗](#)
10. R-TOFU: Unlearning in Large Reasoning Models
Sangyeon Yoon, Wonje Jeung, Albert No
Conference on Empirical Methods in Natural Language Processing (EMNLP 2025 Main) [\[Link\]](#) [🔗](#)
11. SEPS: A Separability Measure for Robust Unlearning in LLMs
Wonje Jeung*, **Sangyeon Yoon***, Albert No
Conference on Empirical Methods in Natural Language Processing (EMNLP 2025 Main) [\[Link\]](#) [🔗](#)
12. Adversarial Sample-Based Privacy Auditing
Sangyeon Yoon*, Wonje Jeung*, Albert No
Workshop on Statistical Foundations of LLMs and Foundation Models (SFLLM@NeurIPS 2024) [\[Link\]](#) [🔗](#)

Tech Report

1. EXAONE 4.5 Technical Report: LG’s First Open-Weight Vision-Language Model for Industrial Intelligence
LG AI Research; contributed to build synthetic post-training data
2026 [\[Link\]](#) [🔗](#) [\[Model\]](#) [🔗](#)
2. K-EXAONE Technical Report: Journey to Frontier-Level Performance of Foundation Models
LG AI Research; contributed to build synthetic post-training data
2026 [\[Link\]](#) [🔗](#) [\[Model\]](#) [🔗](#) *Hugging Face #1 Paper of the Day*

Honors and Scholarship

Academy Research Fellowship, BK21

2026

Awarded for the research paper (\$1.5K).

Undergraduate Scholarship

2019 – 2025

Full scholarship (\$58K over 4 years).